

User, Market & Competitor Research

Research objective

- Understand who has the problem, how they currently solve it, why existing approaches are insufficient, and what a sovereign AI workbench must actually provide to be useful.
- The research must establish why the proposed product needs to exist.

Research method

For every major finding:

1. Identify the **specific user/organization** concerned.
2. Identify the **actual workflow** being performed.
3. Identify the **problem or constraint**.
4. Identify how the organization currently handles it.
5. Identify where the current approach fails.
6. Determine whether a sovereign AI workbench could address that failure.
7. Record the source.
8. Distinguish **fact from interpretation**.

Must investigate

- Confidential enterprise knowledge-work workflows
- Engineering document workflows
- Inspection-report workflows
- Approval-note preparation
- Internal knowledge search
- Engineering drawing/P&ID workflows
- Internal software/code workflows
- Enterprise AI adoption barriers
- Confidential-data restrictions
- On-premise AI requirements
- Air-gapped environment requirements
- User expectations from enterprise AI assistants
- Trust requirements
- Auditability expectations
- Human approval requirements
- Acceptance of agentic/autonomous execution
- Existing solutions used by target organizations
- Direct sovereign/on-premise AI competitors

- Indirect alternatives such as manual workflows and cloud AI
- Competitor capabilities
- Competitor limitations
- Competitor deployment models
- Competitor pricing/licensing where publicly available

Keywords

R1 — User Research

- confidential enterprise knowledge work workflows
- confidential document handling workflows
- engineering document review workflow
- inspection report review workflow
- approval note preparation workflow
- engineering drawing review workflow
- internal knowledge search workflow
- enterprise technical knowledge workers
- refinery engineer AI workflow
- PSU engineer AI workflow
- government technical office AI workflow
- defence manufacturing knowledge work
- enterprise employee AI adoption barriers
- confidential data AI usage barriers
- employee use of public AI for confidential work
- enterprise AI assistant expectations
- on-premise AI user expectations
- self-hosted AI assistant user experience
- AI-assisted engineering documentation
- AI-assisted technical report preparation
- AI-assisted internal correspondence
- manual enterprise knowledge-work bottlenecks
- enterprise document processing pain points
- engineering document analysis pain points
- AI trust in confidential enterprise environments
- explainability expectations for enterprise AI
- auditability expectations for enterprise AI
- human approval requirements for AI-generated business artifacts
- user acceptance of autonomous AI agents
- enterprise concerns about AI agents accessing internal files

R2 — Market Research

- sovereign AI enterprise market
- on-premise generative AI market
- private generative AI enterprise adoption
- air-gapped AI enterprise adoption
- enterprise AI for confidential data
- private AI infrastructure adoption
- enterprise local LLM adoption
- industrial generative AI adoption
- AI adoption in refinery engineering
- AI adoption in oil and gas technical operations
- AI adoption in PSU organizations
- AI adoption in defence manufacturing
- AI adoption in government knowledge work
- enterprise multimodal AI adoption
- enterprise AI agent adoption
- enterprise RAG adoption
- enterprise document intelligence adoption
- enterprise AI infrastructure constraints
- on-premise AI deployment constraints
- air-gapped AI deployment constraints
- enterprise GPU availability for local AI
- enterprise AI data sovereignty requirements
- barriers to enterprise generative AI adoption
- barriers to on-premise LLM adoption
- confidential-data AI market requirements

R3 — Competitor Research

- sovereign AI workbench competitors
- self-hosted enterprise AI assistant competitors
- on-premise enterprise AI platforms
- air-gapped enterprise AI platforms
- private AI assistant platforms
- enterprise local LLM platforms
- enterprise agentic AI platforms on-premise
- enterprise RAG platforms on-premise
- local multimodal enterprise AI platforms
- AI platforms for confidential engineering documents
- AI platforms for engineering knowledge management
- AI platforms for industrial knowledge work
- AI coding agents on-premise
- private AI coding assistants

- enterprise AI document agents
- enterprise AI agent platforms
- sovereign AI platform feature comparison
- local AI platform feature comparison
- on-premise AI pricing models
- enterprise AI licensing models
- self-hosted AI licensing
- enterprise AI deployment models
- competitor model routing capabilities
- competitor agentic execution capabilities
- competitor sandboxed code execution
- competitor multimodal document processing
- competitor local knowledge retrieval
- competitor artifact generation
- competitor audit logging
- competitor network isolation
- competitor air-gap support
- competitor hardware requirements

Required Results

Produce:

User/organization

↓

Actual workflow

↓

Pain point

↓

Current solution

↓

Failure/limitation

↓

Required capability

↓

Evidence/source